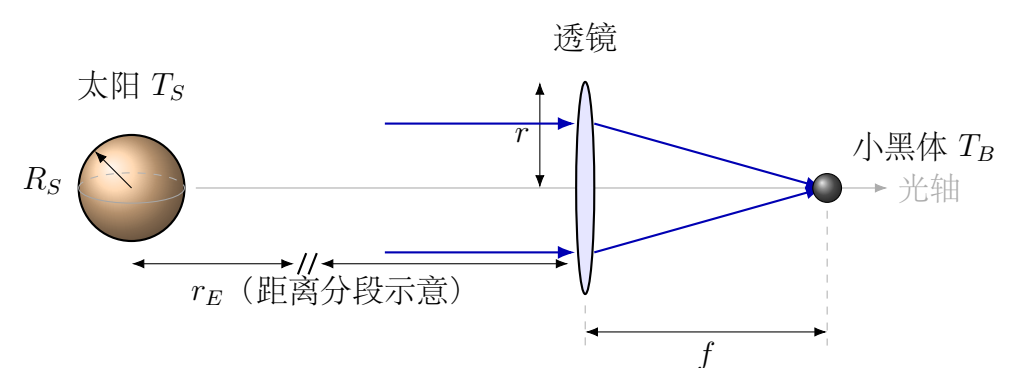
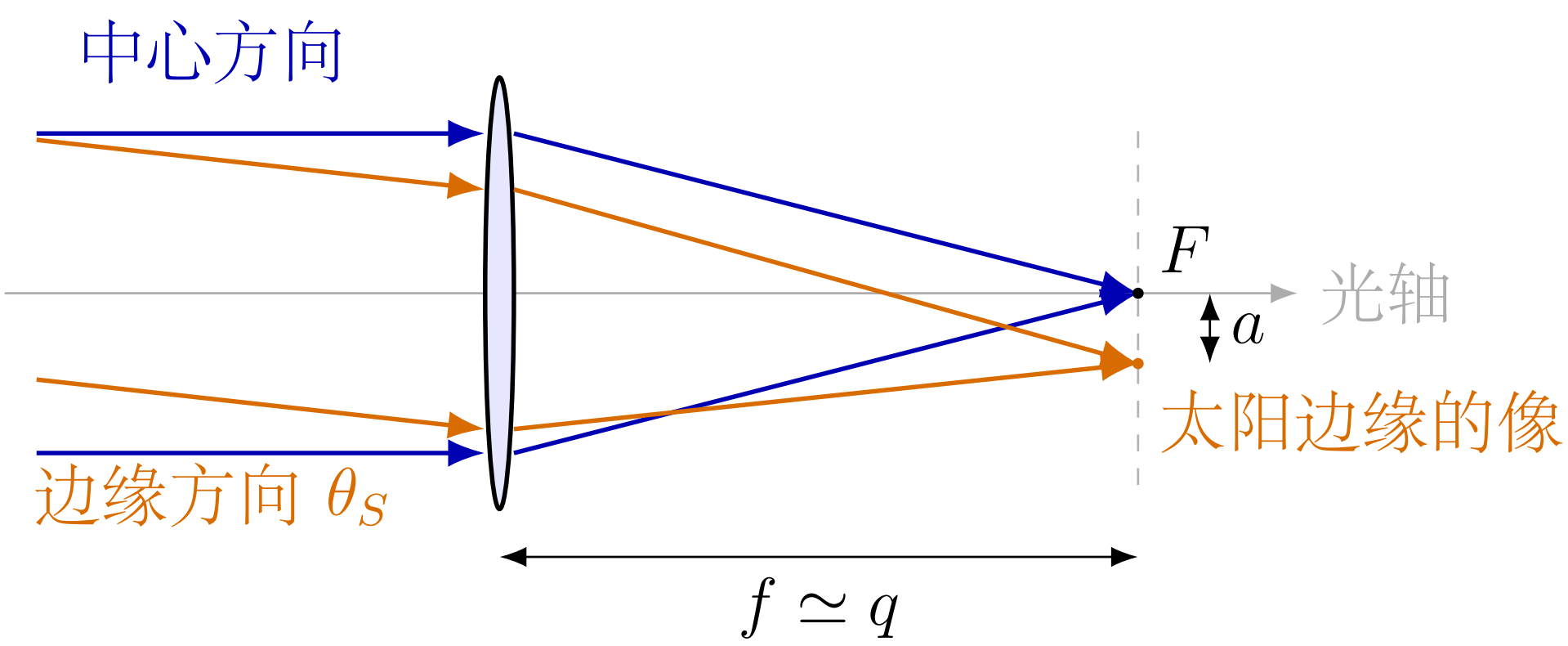
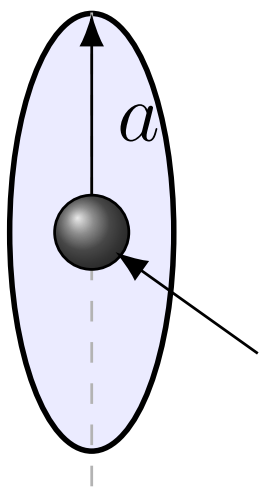

透镜
太阳 T_S
R_S
r
小黑体 T_B
光轴
r_E（距离分段示意）
f

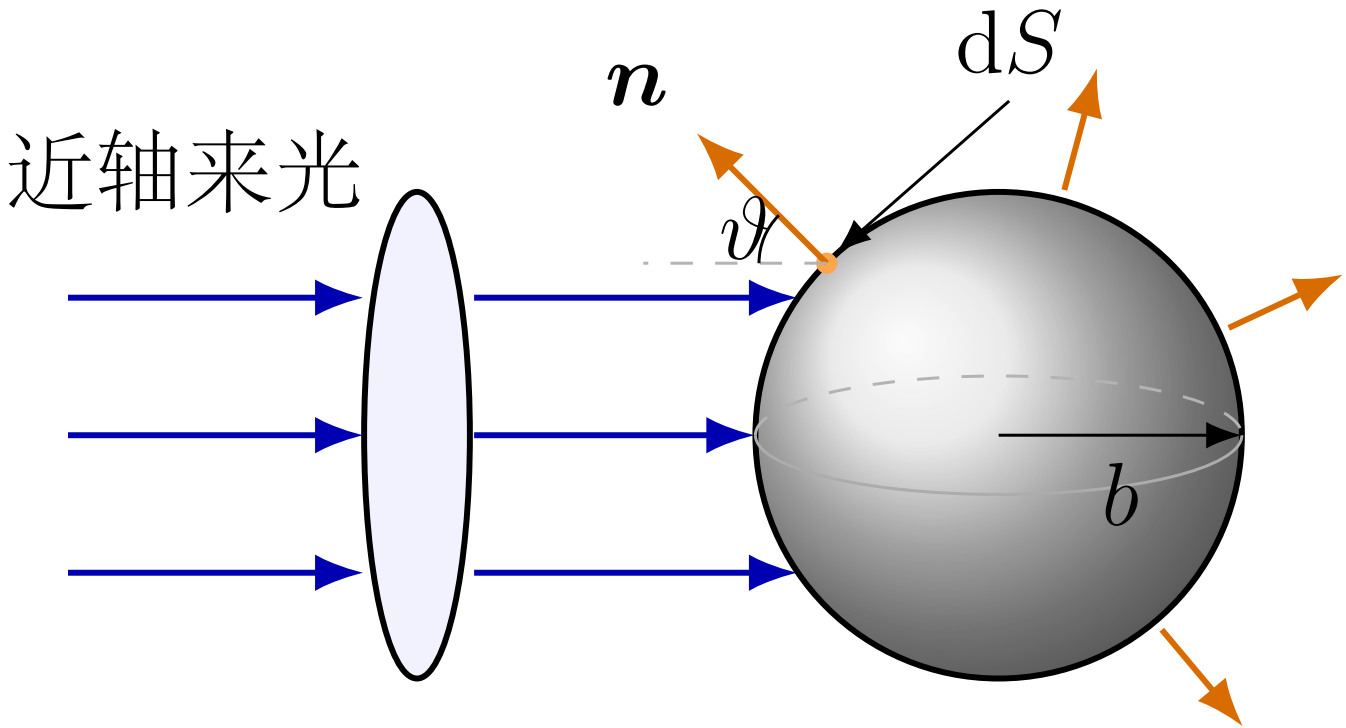


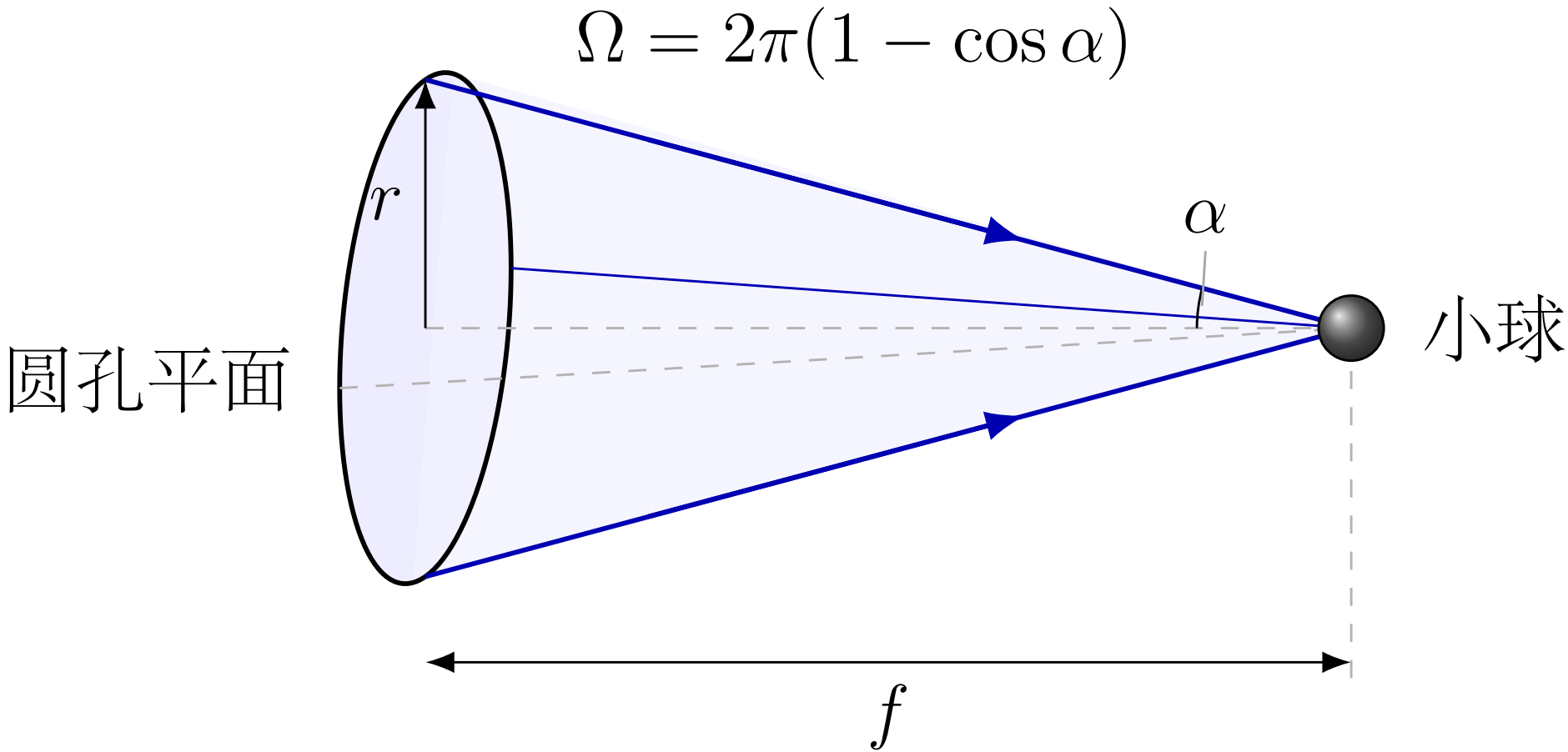
平面圆像

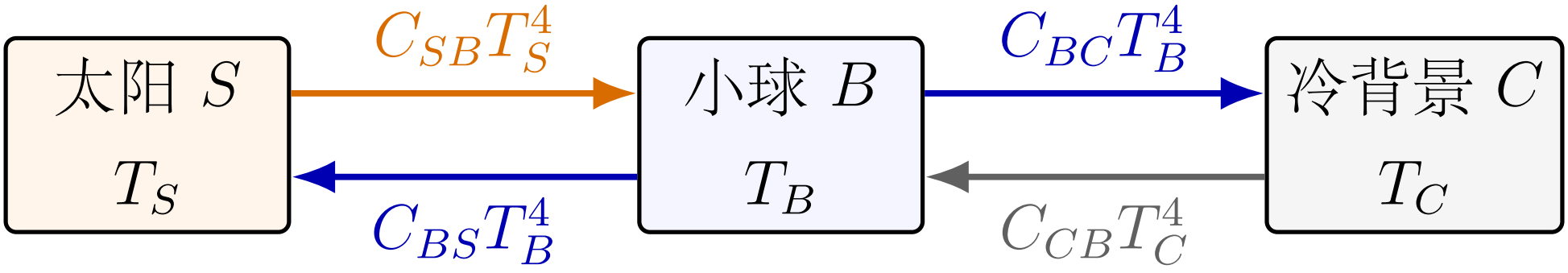
$A_{\rm im} = \pi a^2$

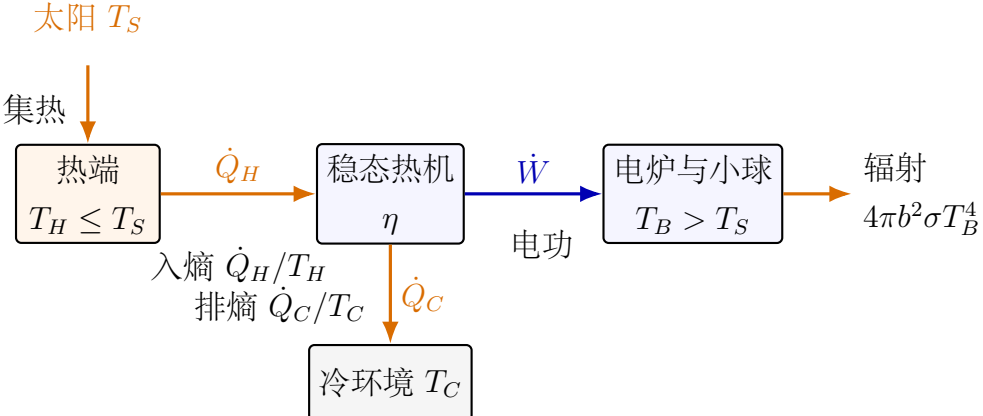


小球半径 $b \ll a$


近轴来光
$\boldsymbol{n}$
$\mathrm{d}S$
ϑ
b
投影 πb^2
等温球面
散热 $4\pi b^2$


$\Omega = 2\pi(1 - \cos\alpha)$
圆孔平面
r
α
小球
f


太阳 S
T_S
小球 B
T_B
冷背景 C
T_C
$C_{SB}T_S^4$
$C_{BS}T_B^4$
$C_{BC}T_B^4$
$C_{CB}T_C^4$
$C_{SB} = C_{BS}$
$C_{BC} = C_{CB}$


太阳 T_S
集热
热端
$T_H \le T_S$
$\dot{Q}_H$
稳态热机
η
$\dot{W}$
电功
电炉与小球
$T_B > T_S$
辐射
$4\pi b^2 \sigma T_B^4$
入熵 $\dot{Q}_H/T_H$
排熵 $\dot{Q}_C/T_C$
$\dot{Q}_C$
冷环境 T_C